

Endoscopy and ERCP in Paediatric Hepatology.

Picture a boy of fourteen months who had a Kasai operation for biliary atresia at eight weeks of age. The operation did not fully clear his jaundice. His spleen is now easy to feel, his platelet count has drifted down to ninety thousand, and his albumin is three point two grams per decilitre. His parents ask a simple question. Is he going to bleed? And a second question follows from the first. Should someone put a camera into his stomach, under general anaesthesia, to find out?

Now picture a different child, a girl of seven weeks with pale stools and a conjugated bilirubin of five milligrams per decilitre. The ultrasound is not conclusive, the liver biopsy is suggestive but not certain, and the magnetic resonance scan of the bile ducts did not show a common bile duct. The surgeon is ready to operate. Is there a test that could prove her bile ducts are open and spare her an operation?

This chapter is about the procedures that answer these questions: upper endoscopy to look for and treat varices, and the two ways of filling the bile ducts with contrast, either from below through the duodenum, or from above through the skin and the liver.

The main idea of the chapter is this. In paediatric hepatology an endoscopy or a cholangiogram is justified only when its result will change what we do next. The procedures are safe in expert hands, but they need general anaesthesia, equipment sized to the child, and a team that does them often. And the evidence behind many of our habits, especially around preventing a first variceal bleed, is much thinner in children than it is in adults. So for each procedure we will ask three things. What does it find? How good is the evidence that finding it helps the child? And when does it actually change management?

A short note on the sources. For the technique of endoscopy in children, the main document is the joint guideline of the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, and the European Society of Gastrointestinal Endoscopy, called ESGE, published in 2017 with Thomson and Tringali as first authors. For portal hypertension in children there are two reports from paediatric meetings held alongside the Baveno consensus conferences, written by Shneider and colleagues in 2012 and 2016, and a recent consensus statement from the Asian Pan-Pacific paediatric society, called APSPGHAN, published in 2026. For adults, which is where most of the evidence lives, we will use the Baveno seven consensus of 2022, the 2024 guidance of the American Association for the Study of Liver Diseases, called AASLD, and the 2018 guideline of the European Association for the Study of the Liver, called EASL, on decompensated cirrhosis. For the cholestatic infant we use the 2017 guideline of the North American and European paediatric societies, NASPGHAN and ESPGHAN, written by Fawaz and colleagues. The

numbers on bleeding come mainly from the Bicêtre hospital group in Paris, led by Duché and later Ackermann, and from the North American research network on childhood liver disease, called ChiLDReN.

A few words first. EGD stands for oesophagogastroduodenoscopy, which is simply upper endoscopy. ERCP stands for endoscopic retrograde cholangiopancreatography. In ERCP a side-viewing endoscope called a duodenoscope is passed into the duodenum, a thin catheter is pushed into the opening of the bile duct, and contrast is injected so that the ducts can be seen on X-ray. PTC stands for percutaneous transhepatic cholangiography. In PTC a radiologist passes a needle through the skin into a bile duct inside the liver and fills the ducts from above. MRCP, magnetic resonance cholangiopancreatography, shows the ducts without any instrument at all. Varices are dilated veins, mainly in the lower oesophagus and the stomach, that open up when blood cannot flow easily through the liver. EHPVO stands for extrahepatic portal vein obstruction, which means the portal vein outside the liver is blocked, usually replaced by a tangle of small collateral veins called a cavernoma. Finally, band ligation means placing small rubber bands around varices through the endoscope, and sclerotherapy means injecting them with a chemical that makes them clot and scar.

Part one. Two kinds of portal hypertension.

Before we decide whether to look for varices, it helps to understand why children with liver disease develop them and why the danger is not the same in every child.

Portal hypertension means raised pressure in the portal vein, the vein that carries blood from the gut and spleen to the liver. When the pressure rises, blood looks for other ways back to the heart. One of those ways runs through veins at the bottom of the oesophagus and the top of the stomach. These veins were never meant to carry much blood, so they swell into varices, and a varix with a thin wall under high pressure can burst.

In children there are two very different ways to get there. The first is disease inside the liver. Biliary atresia is the classic example. After the Kasai operation many children still develop fibrosis and then cirrhosis. The block to flow sits in the sinusoids, the small channels inside the liver, and at the same time the liver gradually loses its ability to make proteins. Albumin falls and the clotting tests worsen. So when a child with cirrhosis bleeds, two things go wrong at once. There is the blood loss itself, and there is the stress of the bleed on a liver that is already failing. A bleed can tip that child into ascites, encephalopathy or infection.

The second way is a block before the liver, in the portal vein itself. This is extrahepatic portal vein obstruction. The liver tissue is usually normal, so synthetic function is normal. These children often present with a first bleed out of nowhere, or with a big spleen and low platelets found by chance. They usually tolerate a bleed well, although bleeding tends to recur. And here is the key point for management. In these children the real target is not the varix but the blocked vein. In many of them a surgeon can build a Meso-Rex bypass, a graft

that carries blood from the mesenteric vein into the left portal vein inside the liver, and restores normal flow. The Baveno six paediatric report and the 2026 Asian Pan-Pacific consensus both say that a Meso-Rex bypass should be done whenever the anatomy allows.

There is one more difference worth knowing. Liver stiffness measured by elastography tracks fibrosis, so in biliary atresia it rises as the liver scars. In portal vein obstruction the liver itself is soft, so liver stiffness can be close to normal even when the varices are large. In that disease the spleen tells you more than the liver.

So the same grade of varix carries a different danger in the two diseases, and the plan after the endoscopy is also different.

Part two. The Paris numbers.

Most of what we know about varices in biliary atresia comes from the Bicêtre hospital near Paris, which serves as a national referral centre. In a study published in *Gastroenterology* in 2010, Duché and colleagues evaluated one hundred and thirty-nine children with biliary atresia. Eighty-eight of them had oesophageal varices, which is about two in every three. About one in five children bled from varices, at a median age of seventeen months. And three quarters of all first bleeds happened before the age of two and a half years.

The most important finding was about what the varices looked like. The children who bled were those with large varices, with red spots on the surface of the varix, and with varices that continued down onto the cardia, the top of the stomach. Endoscopic appearance separated the children at risk better than platelet count alone.

So in biliary atresia, varices and bleeding are mainly a problem of the first three years of life. That is the reason many centres consider a first endoscopy within the first year or two.

Part three. The appearance predicts the bleed.

The Paris group later put this into a definition of high-risk varices. High-risk varices are grade three oesophageal varices, or grade two varices with red signs, or gastric varices running along the cardia. The 2026 Asian Pan-Pacific consensus uses almost the same definition: grade two with red colour signs, grade three, or any gastric varices.

In a large series published in the *Journal of Hepatology* in 2017, covering about thirteen hundred children seen over twenty-five years, Duché and colleagues looked back at who had high-risk varices. Among two hundred and forty-six children who had bled spontaneously, ninety-six percent had high-risk varices. Among eight hundred and seventy-two children who had never bled, only eleven percent did.

A natural history study from the same centre, published in *Gastroenterology* in 2026 with Ackermann as first author, followed more than fifteen hundred children with repeated endoscopies. If a child had high-risk varices and received no prophylaxis, bleeding occurred

in seventy-five percent. If the first endoscopy showed no high-risk varices, bleeding occurred in fewer than four percent. And varices are not static. Among children without high-risk varices at the first endoscopy, thirty-six percent had developed them by five years and fifty-four percent by ten years.

These numbers make a clear case. If you want to prevent a first bleed, the decision rests on what the varix looks like, and that requires an endoscopy.

Part four. North America bleeds less.

There is an important counterweight, and every attending should know it. The Paris data come from a referral centre, and referral centres see sicker children. The ChiLDReN network in North America followed children with biliary atresia prospectively, from the time of diagnosis, in two cohorts. In a paper by Bass and colleagues in *Hepatology* in 2022, eight hundred and sixty-nine children were studied. In the cohort followed from infancy, the five-year incidence of variceal bleeding was nine point four percent. In the cohort of older children still living with their native liver, it was eight percent.

When a bleed did happen, it was rarely fatal. Of eighty children who bled, two died within six weeks, which is two and a half percent, and ten received a transplant within six weeks, which is twelve and a half percent. The children at highest risk were those with an albumin below three point five grams per decilitre, a low platelet count, a high bilirubin, a high aspartate aminotransferase and a high ratio of aspartate aminotransferase to platelets.

The authors concluded that this low risk needs to be weighed when deciding on screening and prophylaxis. Put simply, screening every child with biliary atresia is hard to defend on a five-year bleeding risk of eight to nine percent. Screening the child whose blood tests say that the liver is failing is much easier to defend. The Paris and North American data are not in conflict. They describe different populations, and they tell us to aim screening at the children at highest risk.

Part five. Can we skip the scope?

Because endoscopy in a small child needs a general anaesthetic, people have tried to predict varices with simple measurements. In adults this has worked well. The Baveno seven consensus says that an adult with compensated cirrhosis whose liver stiffness is below twenty kilopascals and whose platelet count is at least one hundred and fifty thousand can safely avoid screening endoscopy. In children the picture is less settled.

The best known paediatric tool is the clinical prediction rule developed in Toronto by Gana and colleagues. It combines the platelet count, the spleen size expressed as a z-score for age, and the albumin. In the first study, of fifty-one children published in 2010, it performed very well, with an area under the curve of zero point nine three, a sensitivity of ninety-four percent and a specificity of eighty-one percent. But the first study of a rule always looks

good.

The real test came in a multicentre validation published in *Gastroenterology* in 2011. It included one hundred and eight children from eight centres, and sixty-nine percent of them had varices. The best single index was the ratio of the platelet count to the spleen z-score, with an area under the curve of zero point eight four. The clinical prediction rule reached zero point eight zero, and the platelet count alone zero point seven nine. These are respectable numbers for ruling varices in. The positive predictive value was about zero point eight six. But the negative predictive value was only about zero point six three to zero point six four. In other words, among children whom the rule called low risk, roughly one in three still had varices.

Other tools tell the same story. The King's variceal prediction score, cited in the 2026 Asian Pan-Pacific consensus, has a positive predictive value of about eighty-two percent, a sensitivity of about seventy-two percent and a specificity of about seventy-three percent. When the adult Baveno six criteria were tested in thirty-three children with cirrhosis, they had a sensitivity of only sixty percent for varices needing treatment. A revised paediatric rule using a liver stiffness below nineteen kilopascals and platelets above one hundred and seventy-five thousand reached one hundred percent in that small group. In a study of forty-eight children with biliary atresia, published in 2024, the Baveno six and seven criteria did identify every child with varices needing treatment. These results are promising, but they are small and single-centre.

So the honest summary is this. No paediatric rule is yet safe enough to cancel an endoscopy on its own. The Asian Pan-Pacific consensus says the same thing: noninvasive tests may help rule disease out, but they do not replace endoscopy. Use the rule to decide which child goes first, and remember that a low score in a child with biliary atresia means less urgency, not no varices.

Part six. Preventing the first bleed: one trial, very low certainty.

Suppose the endoscopy shows high-risk varices. Should we treat them before they bleed? This is called primary prophylaxis. In adults, the options are a non-selective beta-blocker, with carvedilol now preferred, or band ligation. In children the evidence is remarkably thin.

There is exactly one randomised trial of primary prophylaxis in children. It was published by Gonçalves and colleagues in the *Journal of Pediatric Surgery* in 2000. One hundred and eight children with varices were randomised to sclerotherapy or to no treatment and followed for a median of four and a half years. Variceal bleeding occurred in five percent of the sclerotherapy group and forty percent of the control group. Any upper gastrointestinal bleeding occurred in twenty-one percent against forty-six percent. But mortality did not differ, sixteen percent against fifteen percent. And gastric varices developed equally often in both groups, about ten percent.

The Cochrane reviews put this into perspective. The review of sclerotherapy found only this single trial and judged the certainty of the evidence very low. The review of band ligation against sham treatment found only one small feasibility trial and could not draw trustworthy conclusions. The review of beta-blockers against placebo in children found no randomised trials at all.

That is why the paediatric Baveno reports advise against routine beta-blockers for primary prophylaxis in children. Doses, effects and safety have not been established, and the 2026 Asian Pan-Pacific consensus notes that propranolol has limited efficacy and tolerability in children. So every recommendation about primary prophylaxis in children is either an extrapolation from adults or an expert opinion, and it is worth saying that openly to families.

Part seven. What the cohorts add.

Observational studies fill some of the gap. A meta-analysis by Alatas and colleagues, published in 2023, combined four studies with one hundred and seventy-four children. Children who received primary prophylaxis bled in eight point seven percent, while children treated only after a first bleed bled again in thirty-three point seven percent. That corresponds to an odds ratio of zero point two one, although the certainty is low.

In a Brazilian cohort of seventy-two children with portal vein obstruction, published in 2020, medium or large varices carried an eighteen-fold higher risk of bleeding. Primary prophylaxis eradicated the varices in eighty-five point seven percent of children, but they came back in forty-one point seven percent.

In the 2017 Paris series, one hundred and twenty children with cirrhosis and high-risk varices received primary prophylaxis. Ten had a surgical shunt, and none of them bled. One hundred and ten had endoscopic treatment. Seventy-six reached eradication and twenty relapsed. The authors described a fairly good safety record.

The practical message is that banding is a programme, not a single procedure. Once you start, you commit the child to repeat sessions until the varices are gone, and then to surveillance, because varices come back.

Part eight. Size the scope to the child.

The technical side of endoscopy in children decides what is possible. The 2017 ESPGHAN and ESGE guideline gives the key numbers. A paediatric gastroscope is four point nine to six millimetres wide, with a working channel of two millimetres. Diagnostic upper endoscopy is feasible even in neonates of one and a half to two kilograms.

Band ligation is different. The banding devices fit on gastroscopes of eight point five to nine point two millimetres. The guideline makes an important point here. The limit is mainly the narrowness of the child's pharynx, not just the body weight. Traditionally, band ligation was

restricted to children above about eight kilograms, and the 2026 Asian Pan-Pacific consensus prefers band ligation for children of ten kilograms or more. In smaller children, sclerotherapy is the alternative, because a needle can pass through any scope.

When both are possible, band ligation is better. The ESPGHAN and ESGE guideline gives a strong recommendation for band ligation over sclerotherapy, because it eradicates varices more effectively and with less rebleeding.

For ERCP, children below ten kilograms need a paediatric duodenoscope of seven and a half millimetres with a two millimetre channel. This is good for diagnosis but limits what can be done. From ten kilograms a standard therapeutic duodenoscope can be used, which allows stents, stone extraction and dilatation. Both variceal bleeding and ERCP should be managed under general anaesthesia with a secured airway.

Part nine. ERCP in the cholestatic infant.

Let us go back to the seven-week-old girl with pale stools. The question is whether she has biliary atresia, and the answer must come quickly, because the Kasai operation works best when it is done early.

ERCP can show directly whether contrast flows into the bile ducts. If the common bile duct and the ducts inside the liver fill normally, biliary atresia is excluded and the child can avoid an operation. Several centres have published their results. In Prague, Keil and colleagues reported one hundred and four infants in 2010. ERCP had a sensitivity of eighty-six percent and a specificity of ninety-four percent for biliary atresia, and it also identified choledochal cysts accurately. The same group later reported two hundred and fifty-five ERCPs in infants, including one hundred and twenty-one with biliary atresia, with a sensitivity of one hundred percent, a specificity of ninety-three percent and a negative predictive value of one hundred percent. The procedure failed technically in about five percent, and complications occurred in about fourteen percent, but severe complications in fewer than one percent. In Oslo, Åvitsland and Aabakken reported fifty-six infant procedures, with a sensitivity of ninety-five percent and a specificity of eighty-nine percent, and in sixteen of thirty-eight infants with suspected atresia the diagnosis was excluded. A German series from Tübingen, published in 2026, reported sixty infants with a median age of fifty days. ERCP was possible in fifty-eight of them, down to a weight of two point six kilograms, with a sensitivity of one hundred percent and a specificity of ninety-two percent.

Now compare that with MRCP. The 2017 NASPGHAN and ESPGHAN guideline cites a sensitivity of ninety-nine percent for MRCP, but a specificity of only thirty-six percent. Why so low? Because the common bile duct of a cholestatic infant is tiny and often collapsed, so MRCP frequently fails to show it. A duct that is not seen gets read as an absent duct, and many infants without atresia are labelled as having it. ERCP fills the duct with contrast and proves that it is open. In a chart, if you imagine bars for specificity, the MRCP

bar reaches only about a third of the way, while the bars for the four ERCP series all sit near ninety percent.

The guideline still keeps a careful line. It says that MRCP, ERCP and percutaneous cholangiography have a limited role in diagnosing biliary atresia, and that the intraoperative cholangiogram remains the criterion standard. The ESPGHAN and ESGE guideline recommends diagnostic ERCP in infants only when the noninvasive tests are inconclusive. That is the right way to think of it. ERCP is a test for the expert centre, used to prevent an unnecessary laparotomy, and it must never delay a Kasai operation that is needed. But you should never let a non-visualised duct on MRCP decide the question by itself.

Part ten. What ERCP costs a small child.

ERCP is the most complex procedure in this chapter, and its success depends heavily on the size of the child. The Pediatric ERCP Database Initiative, a registry from fifteen North American centres reported by Troendle and colleagues in 2022, included more than eleven hundred procedures. Technical success was ninety point five percent overall, but eighty percent in children aged three years or younger, and seventy-six percent in children under ten kilograms. Adverse events occurred in nine and a half percent.

A systematic review by Usatin and colleagues in 2016, covering two thousand six hundred and twelve children, found an overall complication rate of six percent. Post-ERCP pancreatitis occurred in three percent, bleeding in less than one percent and infection in less than one percent. In infants with neonatal cholestasis, complications occurred in about three percent. A French and Belgian series of two hundred and seventy-one children, published in 2021, found higher numbers, with adverse events in nineteen percent and pancreatitis in twelve percent, and a previous liver transplant increased the risk of sepsis about seven-fold.

The ESPGHAN and ESGE guideline adds three practical points. Paediatric ERCP should be done in high-volume centres with paediatric involvement. Rectal non-steroidal anti-inflammatory drugs to prevent pancreatitis are recommended from the age of fourteen. And radiosensitive organs such as the thyroid and gonads should be shielded. The bottom line is simple. The smaller the child, the lower the success, so infants under ten kilograms belong in a centre that does this regularly.

Part eleven. After transplant, the Roux loop decides the route.

The most common reason for cholangiography in a paediatric hepatology unit is not the newborn but the child after liver transplantation. Biliary complications occur in fifteen to forty percent of paediatric recipients, according to a 2015 review by Feier and colleagues. Anastomotic strictures, at the join between donor and recipient duct, occur in two to thirty-five percent. Non-anastomotic strictures, inside the graft, occur in five to twenty-five percent, and bile leaks in two to fifteen percent.

The route to treat them depends on how the bile duct was reconstructed. In most children the donor bile duct is joined to a loop of jejunum, called a Roux-en-Y hepaticojejunostomy, because the child's own duct is absent or too small, as in biliary atresia. A duodenoscope cannot reach that anastomosis in the usual way. So the route is percutaneous, through the liver. Percutaneous treatment succeeds in about seventy to ninety percent. If the child has a duct-to-duct anastomosis, which is more common in older children and adults, ERCP is possible, with success in eighty to ninety percent, but only fifty-eight to seventy-six percent after living-donor grafts.

Real-life series show what this means for a family. In a Polish series of four hundred hepaticojejunostomies, thirty-five strictures developed, most of them at the anastomosis. The children needed an average of two and a half drainage sessions each. Percutaneous treatment alone succeeded in fifty-seven percent, and with surgery added the outcome was good in about eighty-nine percent. In another series of fourteen children with Roux loops, percutaneous treatment reduced cholangitis from sixty-four percent to seven percent, but with periprocedural complications in about twenty-nine percent.

So before ordering an ERCP after transplant, read the operative note. In most children the duodenoscope cannot reach the anastomosis, and the right referral is to interventional radiology.

Part twelve. When the procedure changes management.

It helps to put all of this into concrete scenarios, because that is where the procedures earn their place.

A cholestatic infant whose ultrasound and biopsy are equivocal can have ERCP at an expert centre. If the ducts fill, atresia is excluded and the laparotomy is avoided.

A child with biliary atresia after a Kasai operation whose platelets are falling and whose spleen is growing should have a screening endoscopy. If high-risk varices are found, a band ligation programme starts.

A child with portal vein obstruction at diagnosis should have an endoscopy and then imaging of the Rex recess, the part of the left portal vein where a bypass would be connected. The aim is to plan a Meso-Rex bypass.

A child after transplant with a Roux loop who develops cholangitis needs ultrasound and MRCP, and then percutaneous cholangiography with balloon dilatation of the stricture.

An older child with a stone in the common bile duct can have ERCP with sphincterotomy to clear the duct before the gallbladder is removed.

And a child with a bile leak after transplant with a duct-to-duct anastomosis can often have an ERCP stent that seals the leak without surgery.

Part thirteen. What the guidelines say.

When we compare adult and paediatric documents, the differences are large, and some matter for practice.

On screening endoscopy in compensated disease, adult guidelines say avoid it if liver stiffness is below twenty kilopascals and platelets are at least one hundred and fifty thousand. The 2026 Asian Pan-Pacific consensus recommends a first endoscopy at the diagnosis of portal hypertension in cirrhosis, with follow-up every one to three years, and in biliary atresia it suggests a first look at nine to twelve months of age. The ESPGHAN and ESGE guideline makes no recommendation on screening. This is a material difference.

On noninvasive tests, the adult documents accept a liver stiffness of twenty-five kilopascals or more as proof of clinically significant portal hypertension, and the 2024 AASLD guidance adds spleen stiffness to spare more endoscopies. The paediatric documents accept noninvasive tests only for ruling out, based on small studies. The documents differ.

On drug prophylaxis, the difference is material. Adults with clinically significant portal hypertension receive a non-selective beta-blocker, preferably carvedilol, to prevent decompensation. The paediatric reports do not recommend beta-blockers, and there is no paediatric randomised trial.

On endoscopic prophylaxis, the adult EASL guideline offers a beta-blocker or band ligation for medium or large varices. The Asian Pan-Pacific consensus prefers band ligation for high-risk varices in children of ten kilograms or more, with sclerotherapy or beta-blockers as alternatives in smaller children. The documents differ.

On band ligation against sclerotherapy, everyone agrees that band ligation is better. On the definitive treatment of portal vein obstruction, the paediatric documents recommend a Meso-Rex bypass whenever possible, while the adult documents do not address it. That is a gap rather than a disagreement. On the role of ERCP, the documents are aligned. ERCP is mainly therapeutic, used after MRCP, and in infants a diagnostic ERCP is reserved for cases where the noninvasive tests are inconclusive. Finally, on preventing post-ERCP pancreatitis, adults receive rectal non-steroidal anti-inflammatory drugs routinely, while the paediatric guideline recommends them only from the age of fourteen.

Part fourteen. Pearls, and what to do differently in clinic.

Let us return to the boy of fourteen months. His Kasai operation failed, his platelets are ninety thousand and his albumin is three point two. Every marker that predicted bleeding in the ChiLDReN cohort points the wrong way, and he is in the age window where three quarters of first bleeds happen. He should have an endoscopy, and if high-risk varices are found, a band ligation programme, with sclerotherapy if his pharynx is too small for the banding device. He should also be discussed for transplant listing, because his liver is

failing.

Compare a three-year-old with biliary atresia whose bilirubin is normal and whose platelets are two hundred thousand. Her risk of bleeding is low. It is reasonable to defer endoscopy, follow platelets, spleen size and elastography, and look again if they change.

For the seven-week-old girl with a duct that MRCP could not see, the duct has not been proven absent. If an expert ERCP service is available and it will not delay surgery, an ERCP can prove the ducts are open. If not, she needs an operative cholangiogram now.

So here is what to do differently in clinic. First, aim screening endoscopy at the children at highest risk: young children with biliary atresia, falling platelets, a growing spleen and a low albumin. Second, use noninvasive rules to decide who goes first, never to cancel an endoscopy on their own. Third, when you start banding, plan the whole programme, including surveillance. Fourth, in portal vein obstruction, always ask whether a Meso-Rex bypass is possible. Fifth, never let a non-visualised duct on MRCP decide a Kasai operation. Sixth, send infants under ten kilograms who need ERCP to a high-volume centre. And seventh, after transplant, read the operative note before ordering an ERCP.

Key points.

In biliary atresia, about two thirds of children referred to Paris had varices, and three quarters of first bleeds occurred before the age of two and a half years.

High-risk varices are grade three varices, grade two varices with red signs, or gastric varices along the cardia. They were present in ninety-six percent of children who bled and eleven percent of those who never did.

In the prospective North American ChiLDReN cohorts, the five-year risk of variceal bleeding was only eight to nine percent, and low albumin, low platelets and high bilirubin identified the children at risk.

Paediatric prediction rules have a negative predictive value of only about zero point six three, so they help triage but do not replace endoscopy.

The only randomised trial of primary prophylaxis in children showed fewer bleeds with sclerotherapy, five percent against forty percent, but no difference in survival. There is no randomised trial of beta-blockers in children.

Band ligation is preferred over sclerotherapy whenever the scope fits, and the child's pharynx sets the limit.

MRCP is only about thirty-six percent specific for biliary atresia, while ERCP in expert centres is about ninety percent specific and can spare a laparotomy.

ERCP technical success falls to about seventy-six percent in children under ten kilograms.

After transplant with a Roux loop, strictures are treated percutaneously rather than by ERCP.

Questions to test yourself.

Question one. What three endoscopic features defined high-risk varices in the Paris studies?

Answer: grade three oesophageal varices, grade two varices with red signs, and gastric varices extending along the cardia.

Question two. In the 2010 Paris study of biliary atresia, what proportion of first variceal bleeds occurred before the age of two and a half years? Answer: about three quarters.

Question three. What was the five-year incidence of variceal bleeding in the two ChiLDReN cohorts, and why does it differ from the Paris figures? Answer: nine point four percent and eight percent. The ChiLDReN cohorts were followed prospectively from diagnosis, while the Paris data come from a referral centre that sees sicker children.

Question four. In the multicentre validation of the clinical prediction rule, what was the negative predictive value, and what does that mean at the bedside? Answer: about zero point six three. About one in three children called low risk still had varices, so the rule cannot be used to cancel an endoscopy.

Question five. What did the only paediatric randomised trial of primary prophylaxis show?

Answer: sclerotherapy reduced variceal bleeding from forty percent to five percent, but mortality did not change, sixteen percent against fifteen percent.

Question six. Why is MRCP so poorly specific for biliary atresia? Answer: the common bile duct of a cholestatic infant is tiny and often collapsed, so MRCP often fails to show it, and a duct that is not seen is read as absent. Its specificity is about thirty-six percent.

Question seven. What size of duodenoscope is used in a child under ten kilograms, and what is the technical success of ERCP in that group? Answer: a seven and a half millimetre paediatric duodenoscope, and technical success of about seventy-six percent in the North American registry.

Question eight. A child after liver transplant with a Roux-en-Y hepaticojejunostomy develops cholangitis and an anastomotic stricture. Which procedure should you request?

Answer: percutaneous transhepatic cholangiography with balloon dilatation, because a duodenoscope usually cannot reach a Roux-en-Y anastomosis.

End of chapter.